

Validation of Newtonian long-bubble simulations in circular tubes

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Abstract

A long gas bubble translating through a wetting liquid leaves an annular film between its interface and the tube wall. We assess axisymmetric two-phase simulations of this process against the Bretherton asymptote and a finite-capillary-number representation of Taylor’s measurements. Four calculations at one maximum refinement level reproduce the reference flat-film thickness within 2% over $0.0054 \lesssim Ca_b \lesssim 0.068$. The agreement is preliminary because no spatial refinement sequence has yet been completed. We distinguish the flat film from the smaller local gap beneath the rear meniscus and define a grid-sensitivity study using the four existing physical inputs. This study tests whether the measured film thickness and bubble speed are independent of spatial resolution before extending the capillary-number range.

1 Physical problem and reference limits

Viscous stress draws liquid into the film around a translating bubble, while surface tension connects the nearly cylindrical film to the curved front and rear menisci. Let R_t denote the tube radius, R_v the volume-equivalent bubble radius, U_b the bubble speed, and $\bar{U}$ the imposed mean liquid speed. We define

$$H = \frac{b_{\text{flat}}}{R_t}, \quad Ca_b = \frac{\mu_c U_b}{\sigma}, \quad Ca_{\text{in}} = \frac{\mu_c \bar{U}}{\sigma}, \quad (1)$$

where μ_c is the continuous-phase viscosity and σ is the surface tension. The simulations prescribe Ca_{in} , whereas the motion of the front tip determines Ca_b . The reference laws must therefore be evaluated at the measured bubble speed.

For a long inviscid bubble at small capillary and Reynolds numbers, Bretherton (1961) matched the viscous dynamic meniscus to the capillary outer shape and obtained

$$H_B \simeq 0.643(3Ca_b)^{2/3} \simeq 1.337Ca_b^{2/3}. \quad (2)$$

The result also assumes negligible gravity, a perfectly wetting liquid, and separation between the two dynamic menisci. The rounded coefficient 1.34 is conventional, but the unrounded value matters when errors below one per cent are discussed. A limiting-case test must show that $H/Ca_b^{2/3} \rightarrow 1.337$ as $Ca_b \rightarrow 0$; a line with slope 2/3 over a moderate range does not establish this limit.

Bretherton’s second observable is the relative speed excess,

$$W = \frac{U_b - \bar{U}}{U_b} \simeq 1.29(3Ca_b)^{2/3}. \quad (3)$$

If the uniform film is stationary in the laboratory frame, volume conservation gives

$$\frac{\bar{U}}{U_b} = (1 - H)^2, \quad W = 1 - (1 - H)^2. \quad (4)$$

The liquid flux in the simulated film must be measured before using Eq. (4) as an exact identity. Finite gas viscosity, meniscus interaction, and flow within the film can all modify it.

Pressure comparisons require a stated sign convention. Taking the positive excess pressure from the liquid behind the bubble to the liquid ahead of it, the summed front-and-rear dynamic contribution is

$$\Delta p_{\text{bubble}} \simeq 4.52 \frac{\sigma}{R_t} (3Ca_b)^{2/3}. \quad (5)$$

The coefficient 3.58 quoted in the abstract of Bretherton (1961) is the leading-meniscus dynamic correction. It should not replace 4.52 in a comparison across a complete long bubble. Numerical pressures must be sampled outside both dynamic menisci using the same sign and radius convention.

At finite capillary number, Aussillous and Quéré (2000) represented the circular-tube measurements of Taylor (1961) by

$$H_T = \frac{1.34Ca_b^{2/3}}{1 + 3.35Ca_b^{2/3}}. \quad (6)$$

This rational expression is an empirical fit introduced by Aussillous and Quéré, rather than an equation proposed by Taylor. It approaches Eq. (2) at small Ca_b and accounts empirically for geometrical confinement as the film thickens. Taylor measured the deposited liquid fraction m behind an advancing gas finger; for a uniform annular film, $m = 1 - (1 - H)^2$. Aussillous and Quéré’s high-speed experiment instead moved a finite liquid column with air. Its inertial thresholds Ca^* and Ca^{**} are consequently not universal transition values for the finite gas bubbles considered here.

2 Numerical method and provenance

The calculation solves the axisymmetric, incompressible two-phase Navier–Stokes equations. A volume-of-fluid field tracks the gas–liquid interface, surface tension supplies the capillary stress, and an embedded boundary represents the circular tube wall. The no-slip wall is fully wetted by the continuous phase. A Poiseuille profile drives the liquid at the inlet, and the bubble translates without touching the wall.

Surface tension, μ_c , and R_v define the repeating scales. The present calculations use

$$\frac{R_t}{R_v} = 0.7, \quad \frac{\mu_g}{\mu_c} = 0.01, \quad \frac{\rho_g}{\rho_c} = 0.001, \quad (7)$$

and the volume-radius Laplace number

$$La_v = \frac{\rho_c \sigma R_v}{\mu_c^2} = 1. \quad (8)$$

The tube-based quantities are therefore

$$La_t = 0.7La_v, \quad Re_t = La_t Ca_b, \quad We_t = La_t Ca_b^2. \quad (9)$$

At $Ca_b = 1$, both Re_t and We_t are 0.7. Inertia must therefore be measured during the extension to order-one capillary number, even though both phases remain Newtonian.

The solver and numerical checks use the project-pinned Basilisk release v2026-08-30. A reproducible calculation records this version, the solver source, dimensionless inputs, compiler options, and analysis definitions. The current results use a domain of length $16R_v$ and maximum refinement level 10. The smallest available cell is then $\Delta_{\min} = 16R_v/2^{10} = 0.015625R_v$.

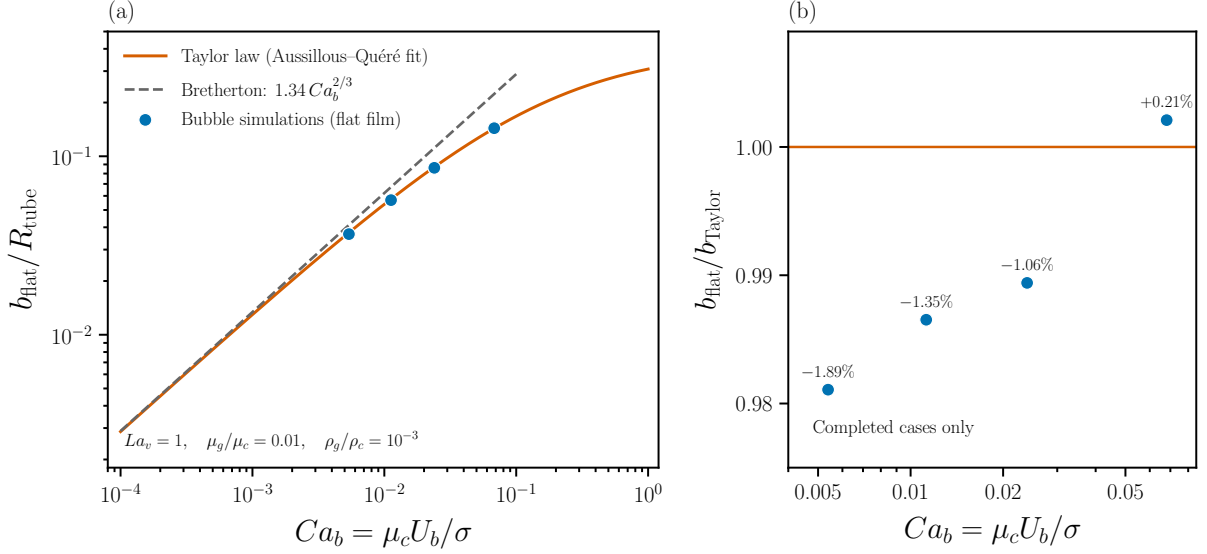


Figure 1: Flat-film thickness against the measured bubble capillary number. Symbols show the four simulations, the solid curve is the Aussillous–Quééré representation of Taylor’s data, and the dashed curve is the small-capillary-number Bretherton asymptote. The second panel shows the ratio to the finite-capillary-number correlation. The reference curves extend beyond the range of completed simulations.

Table 1: Measured flat-film thickness and finite-capillary-number reference values. All four simulations use maximum refinement level 10.

Ca_{in}	Ca_b	H	H_T	$100(H/H_T - 1)$
0.005	0.00538868	0.0366354	0.0373420	−1.892%
0.010	0.0112414	0.0567910	0.0575664	−1.347%
0.020	0.0239374	0.0861486	0.0870712	−1.060%
0.050	0.0680939	0.143663	0.143363	+0.209%

3 Preliminary flat-film comparison

Figure 1 compares the four completed calculations with Eqs. (2) and (6). The plotted data are also supplied in `data/baseline-film.csv`.

The flat-film estimator removes 30% of the bubble’s axial extent at each end, takes the median interface radius within the remaining central region, and averages the result over the final quarter of the usable snapshots. A linear fit to the front-tip position over the same temporal window gives U_b . This procedure is designed to exclude both dynamic menisci. The existence of a spatial plateau and sensitivity to the trimming and averaging windows have not yet been quantified.

The global minimum gap near the rear meniscus is smaller. Its deviations from H_T are approximately −32.3%, −25.9%, −23.7%, and −18.4% for the four cases in Table 1. These values describe a different observable: rear capillary waves can produce a local minimum below the uniform film, as found in the rear-meniscus calculations of Giavedoni and Saita (1999). A steady minimum gap does not by itself establish a steady flat film, pressure drop, bubble length, or complete travelling shape.

At level 10, the measured flat films span at most 1.64, 2.54, 3.86, and 6.44 smallest cells. These are geometric estimates, rather than verified counts of the adaptive leaves within each film. No case presently has a completed refinement sequence. The observed agreement within 2% is therefore a single-resolution comparison and cannot yet be interpreted as numerical accuracy.

Table 2: Grid-sensitivity matrix. “Pending” denotes a calculation under investigation and carries no implied result.

Ca_{in}	Level 9	Level 10	Level 11	Level 12	Finer
0.005	pending	existing	pending	pending	conditional
0.010	pending	existing	pending	pending	conditional
0.020	pending	existing	pending	pending	conditional
0.050	pending	existing	pending	pending	conditional

4 Grid sensitivity and extension in capillary number

The first grid study holds the four existing input values $Ca_{\text{in}} = 0.005, 0.01, 0.02$, and 0.05 fixed. Each physical input is examined at maximum levels 9, 10, 11, and 12, where level 10 is the existing calculation. Selected low-capillary-number cases may require finer stages if these four levels do not reach an asymptotic range. Table 2 records the design; only its level-10 column contains results.

The same inlet capillary number and physical properties are retained across a grid sequence. Both H and the resulting Ca_b must converge; retuning the inlet speed would hide part of the numerical response. The correlation is evaluated at the measured Ca_b on every grid. Bubble volume, length, flat-film uniformity, and temporal drift accompany the two primary observables. At least three solutions in an asymptotic range are required before assigning a convergence order. Time-step and pressure-solver tolerances, domain length, boundary clearance, bubble length, and gas viscosity ratio are separate tests and should not be folded into the spatial sequence.

Adaptive refinement currently responds to interface, curvature, velocity, and strain-rate indicators. A feature-based criterion that guarantees a minimum number of cells across the wall–interface gap may reduce the cost of finer stages. This remains a proposed numerical method. Its feature definition, refinement floor, spatial band, interaction with the embedded boundary, and volume-conservation tests must be fixed before it can contribute evidence to the grid study.

Once spatial sensitivity is known, the bubble calculations can extend from the small- Ca_b limit to $Ca_b = 1$. Equation (6) supplies a convenient film-thickness comparison but does not replace independent benchmarks. The axisymmetric Stokes solutions of Reinelt and Saffman (1985) provide deposited fraction, finger radius, nose shape, and front pressure response at finite Ca_b . Giavedoni and Saita (1997) and Giavedoni and Saita (1999) give complementary front and rear reference problems, including streamline topology and rear interfacial undulations. Their plane-channel results must not be mixed with the circular-tube geometry.

5 Present conclusion

The present calculations support one bounded statement. Four level-10 simulations of Newtonian gas bubbles reproduce the flat-film value of Eq. (6) within 2% over $0.0054 \lesssim Ca_b \lesssim 0.068$. The minimum gap beneath the rear meniscus does not test the same quantity. Spatial convergence, the limiting Bretherton coefficient, speed excess, meniscus profiles, and pressure response remain unestablished. The pending refinement matrix is designed to quantify the first of these uncertainties before the comparison is extended through $Ca_b = 1$.

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